

TIA–FPGA – FPGA replacement for the Atari 2600 Television Interface Adaptor (C010444)

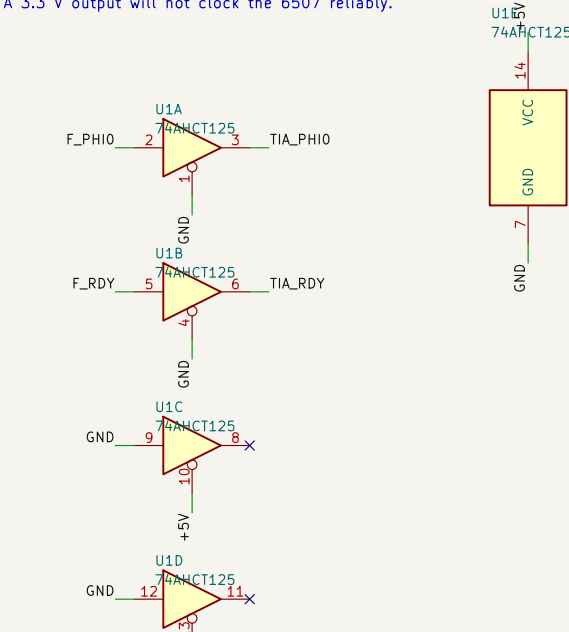
Rev A – digital interface only. Chroma and paddles do not fit the pin budget; see the notes.

FPGA MODULE – Sipeed Tang Nano 9K (GW1NR–LV9QN88PC6/I5)

Pin map verified against both Sipeed board revisions (3672 and 3674), which agree. Of the 48 header pins only 29 are usable 3.3 V GPIO: 8 are BANK3 at 1.8 V, 8 are HDMI differential pairs, 3 are power. Rev A uses all 29, with none spare.

LEVEL TRANSLATORS

U1 is the critical part: the NMOS 6507 requires VIH(PHI0) = Vcc–0.2 V ~ 4.8 V. A 3.3 V output will not clock the 6507 reliably.

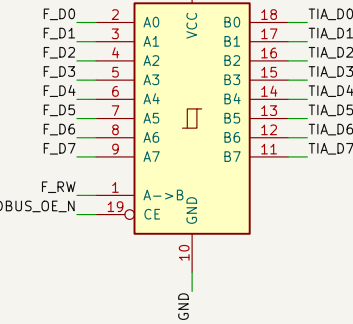


U1 runs from +5 V – this is where the level step–up happens. Spare buffers 3 and 4: /OE tied to +5 V, input tied to GND.

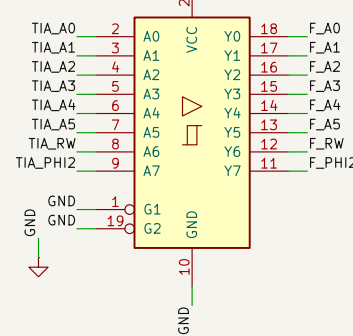
J4/2–9 are BANK3 at 1.8 V – unusable at 3.3 V without a local LDO and a 1.8 V–powered buffer. Recovering those 8 pins is the rev B fix.

J4/18 (+5 V from the Tang Nano) left open on purpose: this board is fed from the breadboard rail, so tying both would put two sources on one net. Power the Tang Nano over USB. +3V3 is taken from J4/24.

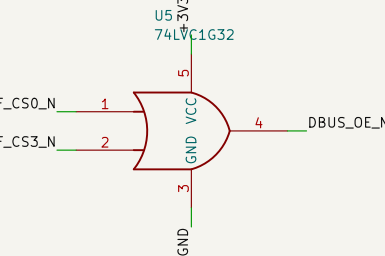
U2 D0–D7. A side = FPGA, B side = TIA, so DIR can be driven straight from the buffered R/W: DIR = (read) sends FPGA -> TIA. That, plus the U5 gate, keeps both 245 control lines off the FPGA – saving the one pin the header budget was short by.



U3 A0–A5, R/W and PHI2 – permanently enabled (G1=G2=GND).

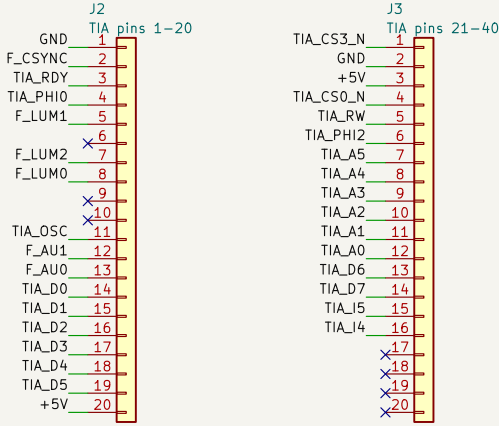


U5 /OE = /CS0 OR /CS3. The 245 drives the bus only while the TIA is selected, so it never fights the ROM or the RIOT on a CPU read.



TIA SOCKET – DIP–40

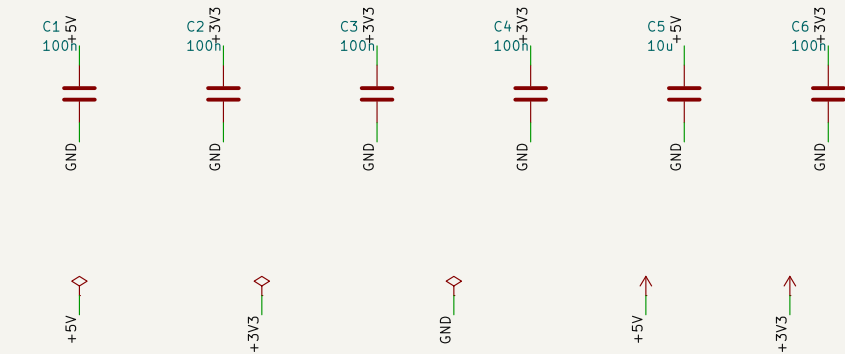
Two 20–way SIP strips that plug into the 2600 socket or a breadboard, in place of the C010444.



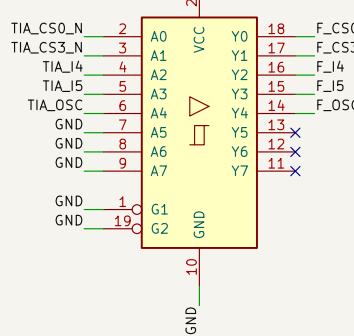
Pin 9 (COL, chroma) and pins 37–40 (paddles) left open: no FPGA pin is left for them on a Tang Nano 9K. See README.

POWER AND DECOUPLING

+3V3 comes from the Tang Nano (J4/24). +5 V comes from the breadboard rail.



U4 /CS0, /CS3, triggers and the 3.579545 MHz clock from the crystal.



VIDEO OUTPUT – CX–2600A resistor network (phase 6, not populated)

4:2:1 luma weighting. Output stage (2N3563) follows the original sheet.

